

Assignment No 10

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Question:

A10. Write a program that handles runtime errors such as division by zero and invalid array index access using built-in exception handling mechanisms. In addition, define and implement custom exception classes to manage application-specific errors such as invalid input values or operation limits, ensuring clear error messages and safe program execution.

Code:

```
/*
A10. Write a program that handles runtime errors such as division by zero and
invalid array index access using built-in exception handling mechanisms.
In addition, define and implement custom exception classes to manage
application-specific errors such as invalid input values or operation limits,
ensuring clear error messages and safe program execution.
*/

#include <iostream>
#include <exception>
using namespace std;

class InvalidInputException : public exception {
public:
    const char* what() const noexcept override {
        return "Invalid input value!";
    }
};

class LimitExceededException : public exception {
public:
    const char* what() const noexcept override {
        return "Operation limit exceeded!";
    }
};

int main() {
    try {
        int a, b;
        cout << "Enter two numbers for division: ";
        cin >> a >> b;

        if (b == 0)
            throw runtime_error("Division by zero error!");

        cout << "Division Result: " << a / b << endl;

        int arr[5] = {10, 20, 30, 40, 50};
        int index;

        cout << "Enter array index (0-4): ";
        cin >> index;

        if (index < 0 || index >= 5)
```

```

        throw out_of_range("Invalid array index!");

    cout << "Array Value: " << arr[index] << endl;

    int input;
    cout << "Enter a positive number: ";
    cin >> input;

    if (input < 0)
        throw InvalidInputException();

    int value;
    cout << "Enter value (limit = 100): ";
    cin >> value;

    if (value > 100)
        throw LimitExceededException();

    cout << "Program executed successfully!" << endl;
}

catch (const out_of_range& e) {
    cout << "Array Error: " << e.what() << endl;
}

catch (const runtime_error& e) {
    cout << "Runtime Error: " << e.what() << endl;
}

catch (const exception& e) {
    cout << "Exception: " << e.what() << endl;
}

return 0;
}

```

Output:

```
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 10> .\a.exe
Enter two numbers for division: 2
0
Runtime Error: Division by zero error!
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 10> .\a.exe
Enter two numbers for division: 1
2
Division Result: 0
Enter array index (0-4): 6
Array Error: Invalid array index!
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 10> .\a.exe
Enter two numbers for division: 1
2
Division Result: 0
Enter array index (0-4): 3
Array Value: 40
Enter a positive number: 2
Enter value (limit = 100): 60
Program executed successfully!
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 10> █
```